

Bus MP13P1993 — Profitability & Revenue Analysis

Stack: Google Sheets → SQL Server → Power BI **Period:** January 2026 – June 2026

1. Problem Statement

Evaluate whether Bus MP13P1993 is holding its profitability over time, and — where profitability is slipping — determine whether the cause is falling revenue, rising costs, or reporting discrepancies. The analysis supports real operating decisions for a family-run transportation business, so findings are presented in plain, non-technical language.

2. Data

Attribute	Detail
Source	Daily manually-entered records in Google Sheets
Granularity	Daily
Range	Jan 2026 – June 2026

Fields: stand-wise passenger deposits (5 outbound stops + 3 return legs + occasional Badnagar leg), luggage revenue, fuel/maintenance/labour/other expenses, in-sheet Gross Income / Total Expenses / Net Income, a Discrepancy field, driver/conductor names, free-text notes.

Known limitations:

- Non-operating days appear only as missing dates, with no reason recorded. Cannot be reliably reconstructed retroactively.
- Fuel is tracked in ₹, not litres, so true fuel efficiency isn't calculable — Fuel Cost per KM is used as the substitute metric.

- The **Discrepancy** field is mixed text/numeric and mostly reflects small, routine, explainable gaps — excluded from the KPI set for that reason.
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3. Tools & Scope

Tool	Role
Google Sheets	Source data entry; derivation of Passenger_Revenue
SQL Server	Cleaning, validation, staged loading, monthly KPI aggregation
Power BI	Dashboard — 6 KPI cards, trend charts, monthly breakdowns, weekday/weekend comparison, month/year slicers

In scope: single bus (MP13P1993), single route, Jan 2026–present, monthly-level trend analysis across 6 KPIs.

Out of scope: fleet-wide comparison (only 1 of 9 buses digitized), stand-wise/route-leg breakdown (deferred to a future phase), precise causal attribution of non-operating-day revenue loss (data can't support it), true fuel efficiency in litres/distance.

4. Data Model & Cleaning

Raw wide-format sheet data was mapped into a **bus_daily_records** SQL Server table. Stand-wise/return-leg columns were retained in the schema but collapsed into a single derived **PASSENGER_REVENUE** column for the current scope.

Load strategy: staged into an all-VARCHAR table first, then explicitly converted and validated into the typed table using **TRY_CONVERT** and **NULLIF**, so bad or inconsistent values surface visibly instead of failing or corrupting silently.

Data issues encountered and resolved:

1. **Currency formatting broke CSV parsing** — money columns exported as quoted, \$-prefixed values with comma thousand-separators (e.g. "\$6,350.00"), misaligning rows on bulk load. Resolved at the source by reformatting to plain numbers in Sheets before export.
2. **Mixed date formats** — earlier entries in DD/MM/YYYY, later entries in DD-MM-YY. Resolved with a COALESCE of two TRY_CONVERT attempts per row.

Query structure: two CTEs — one handling null-value cleanup on the raw table (e.g. COALESCE-ing nullable columns like COLLECTED before use), a second for monthly aggregation via GROUP BY. A self-built arithmetic identity check ($\text{Gross_Income} - \text{Income} - \text{Total_Expenses}$, which must equal 0) is included in the query to catch aggregation errors rather than relying on manual spot-checks.

5. KPIs

KPI	Formula	Purpose
Gross Income	$\text{Passenger_Revenue} + \text{Luggage_Revenue}$	Traffic/demand indicator
Net Profit Margin	$\text{Net Income} \div \text{Gross Income}$	Overall operating efficiency
Gross Margin (ex-fuel)	$(\text{Gross Income} - \text{Fuel Cost}) \div \text{Gross Income}$	Isolates fuel-driven vs. non-fuel profitability drag
Fuel Cost per KM	$\text{Diesel Cost} \div 190$ (fixed daily distance)	Fare-pricing sanity check
Maintenance % of Expenses	$\text{Maintenance Cost} \div \text{Total Expenses}$	Tracks cost-structure shifts, flags repair spikes

Luggage Revenue
%

Luggage Revenue ÷ Gross
Income

Whether luggage is a
meaningful revenue
stream

Discrepancy was evaluated as a potential 7th KPI but excluded — it reflects routine, explainable gaps rather than a systemic reporting issue.

6. Key Findings

Overall profitability: Average Net Profit Margin across Jan–Jun 2026 was ~17%, ranging from ~8% (June) to ~24% (April) — a real, meaningful swing month to month.

Fuel is the dominant driver of margin swings, not maintenance or labour. Gross Margin (ex-fuel) isolates the impact of every cost *except* fuel; the gap between it and Net Profit Margin represents the non-fuel cost burden (maintenance + labour + other) as a share of revenue. This gap stayed nearly constant (~14–16 percentage points) across every month, while Net Profit Margin and Gross Margin (ex-fuel) both swung by a similar ~15–16 points together. A stable gap alongside a large swing in both margin metrics indicates the swing is driven by fuel, not by the non-fuel costs the gap represents.

June's dip is externally verified. Diesel prices across Madhya Pradesh spiked sharply in late May 2026 (confirmed via independent web search) and stayed elevated through June. June's Fuel Cost per KM was the highest of the year, consistent with this external, verifiable event.

May's dip had a different cause: low ridership. Despite the real diesel price spike, May's Fuel Cost per KM was actually the *lowest* of the year. Instead, Gross Income per operating day dropped substantially (~₹4,957/day vs. ~₹6,000–6,400/day baseline) despite full operation (31 days, no stoppage). Confirmed root cause: lower ridership that month.

March's elevated Fuel Cost per KM is not explained by diesel price — prices were essentially flat in March, before the real spike occurred. This points to a consumption-side cause (route conditions, traffic, driving pattern) rather than a price-side one. Root cause remains open.

Operating days and margin are two distinct effects. March and June had fewer operating days (maintenance downtime), which explains lower *total monthly revenue* in those months. This is separate from the fuel-cost findings above, which explain lower *margin/efficiency* — the two should not be conflated.

Weekday vs. weekend: average daily income is ₹5,802 on weekdays vs. ₹5,482 on weekends — the route leans on regular/commuter traffic more than weekend or leisure travel.

Data correction — owner's draw misclassified as an expense. The original sheet's `TOTAL_EXPENSES/NET_INCOME` formulas treated `COLLECTED` (the owner's personal cash draw, a distribution of profit) as if it were a business expense. Corrected during analysis (`Total_Expenses_Adj = Total_Expenses - Collected`, `Income = Net_Income + Collected`), producing a more accurate view of true operating profitability, separate from the owner's personal draw.

7. Open Questions

- **March's elevated Fuel Cost per KM** — consumption-side cause (traffic, route conditions, driving pattern) suspected but not confirmed.
 - **Maintenance's payment timing** — larger repair bills are paid from a separate reserve during the gap period, which explains why big repair events don't spike Maintenance's expense share that month. Whether smaller maintenance bills follow the same deferred pattern as labour costs (paid when cash is available, not necessarily the month the work occurred) is unconfirmed.
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8. Recommendations

For the business:

- Investigate March's fuel cost anomaly directly with drivers/staff — the data has narrowed the question to a consumption-side cause, but the final answer needs operational knowledge the data alone can't supply.
- Consider whether fare or scheduling could better capture weekend demand, given weekdays currently outperform weekends.

For future data collection:

- Add an explicit `Operated (Y/N)` + reason field for non-operating days.
 - Standardize the date entry format in Google Sheets.
 - If feasible, timestamp maintenance and labour costs to the period the work was actually incurred, separate from when it was paid — would resolve the cash-vs-accrual ambiguity in those two KPIs.
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9. Skills Demonstrated

- **SQL Server:** staged data loading, `TRY_CONVERT/NULLIF` type conversion, `COALESCE`-based multi-format date handling, CTEs for null handling and monthly aggregation, self-built data validation checks, debugging an aggregation bug traced to alias-resolution behavior.
- **Power BI:** Power Query (M) data transformation including a manually-built date dimension table, DAX measures, relationship modeling, dashboard design for a non-technical audience.
- **Analytical judgment:** distinguishing volume effects from margin/efficiency effects, identifying and correcting a cash-basis/accrual-basis data conflation, verifying an internal finding against external data (diesel pricing) rather than assuming correlation, and explicitly scoping out claims the data couldn't support.